QC STORY PROJECT

PROJECT TITLE :- To eliminate the V1 and V2 concerns of Tail lamp area damage

Mahindra and Mahindra Ltd, Plant I, Scorpio Body shop.

	Plan	Actual
Start Date	**30/6/12**	**30/6/12**
Completion Date	**28/10/12**	**26/10/12**

Current Status:

Closed

Area of Worry

Damage seen aesthetically before and after paint having V1 severity

Cross Functional Team

No	Description	Name	Designation	Dept	Phone	Email
1	**Team Leader**	Kiran Deshmukh	Asst. Mgr-Prod.	Scorpio Body Shop	8308833228	DESHMUKH.KIRANKUMAR@mahindra.com
2	**Mentor**	R G Suryawanshi	HOD	Scorpio Body Shop	9881378988	
3	**Member 1**	Sanket Pimplapure	Asst. Mgr-MAPs	Scorpio Body Shop	9503500998	
4	**Member 2**	Swapnil Dharrao	Asst. Mgr-Prod.	Scorpio Body Shop	9765813801	
5	**Member 3**	A.R.Shaikh	Exe. Mgr-Tooling	Scorpio Body Shop	9881519976	
6	**Member 4**	Gokul Amrutkar	Asst. Mgr-MQA	Scorpio Body Shop	9922408133	
7	**Member 5**	Rajendra Punde	Asst. Mgr-MTN	Scorpio Body Shop	9922228599	
8	**Member 6**	Pankaj Yelmame	Asst. Mgr - Ding Dent Team	Scorpio Body Shop	9922403871	

1. Select the Theme

Brief description of Problem

What: W area & tail lamp area of the body is damaged

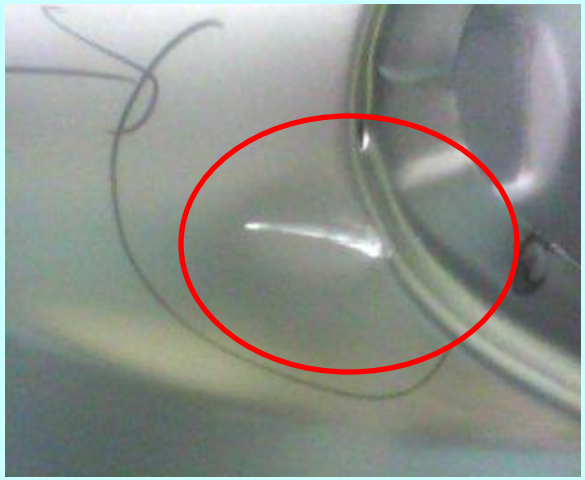
Where: Side Assy LH/RH & Respot #60 on framing line

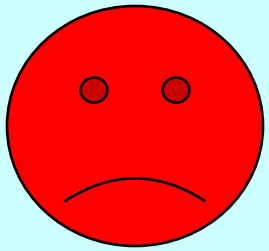
When : Daily(1 & 2 shift)

Who : Side Check man/Final buyoff/CAI

Why : Root cause unknown, high rework cost and time.

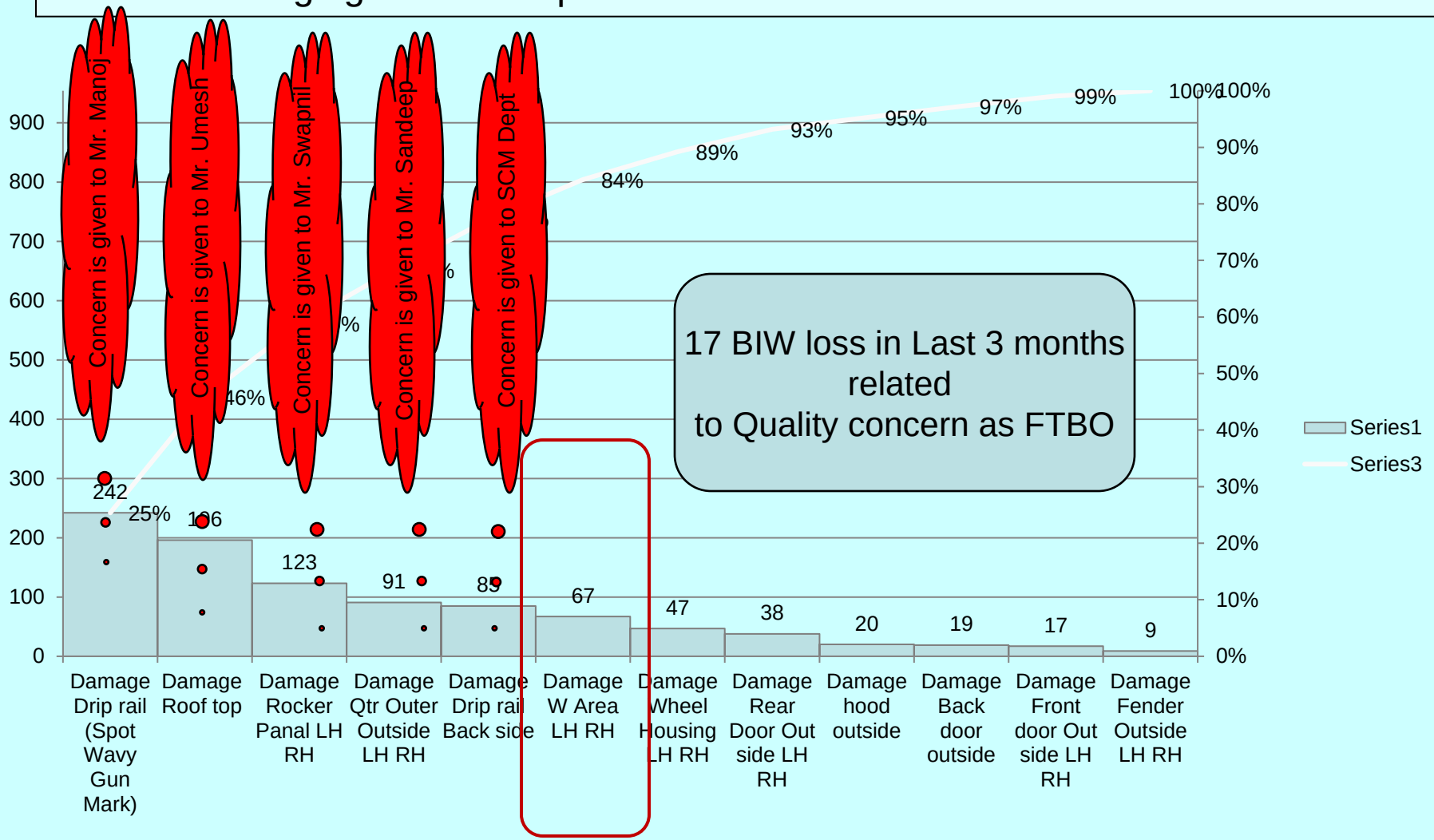
How: 67 incidences having V1 severity reported from last 3 months





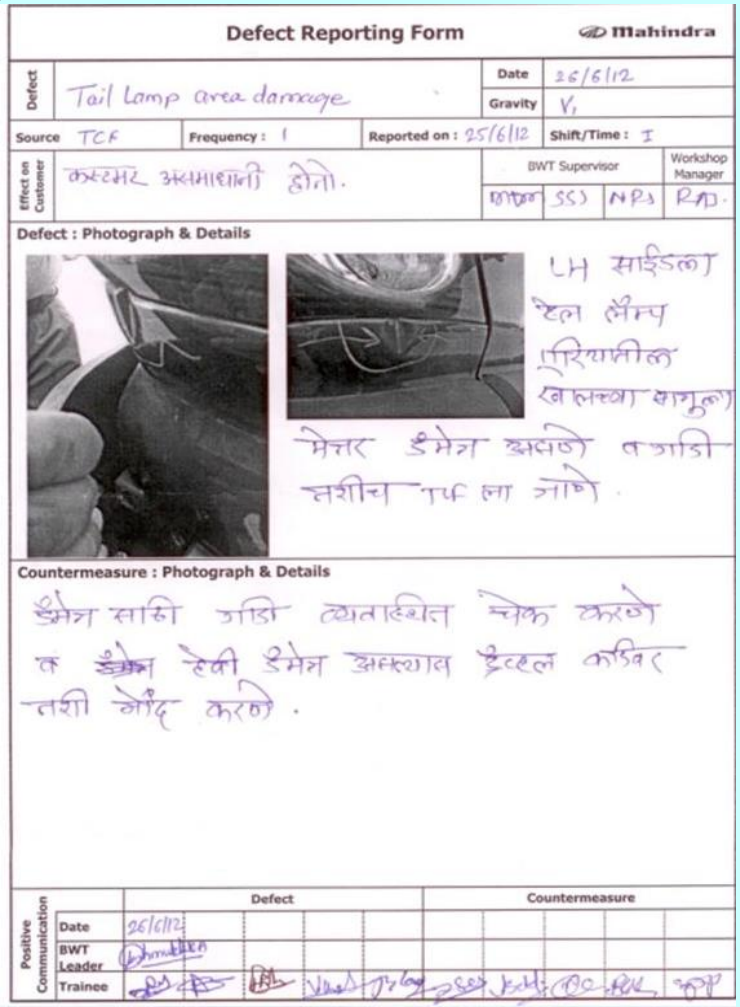
2. Justify the choice (reasons)

Emerging concern reported from TCF on dated 7/5/12 & 25/6/12



2.a Immediate Containment Action

- DRF given to stage operators –If W area & tail lamp area of the body is damaged on stage write on traveler card for rework notification



Defect Reporting Form — Mahindra

Defect	Tail Lamp area damage	Date	26/6/12
		Gravity	V1
Source: TCF	Frequency: 1	Reported on: 25/6/12	Shift/Time: I
Effect on Customer	कस्टमर असमाधानी होतो.	BWT Supervisor	Workshop Manager
		SSJ	NPS / RAD

Defect : Photograph & Details

LH साईडला टेल लैम्प एरियामील खालच्या बाजूला मेटर डैमेज असणे व गाडी तशीच पुढे ला जाणे.

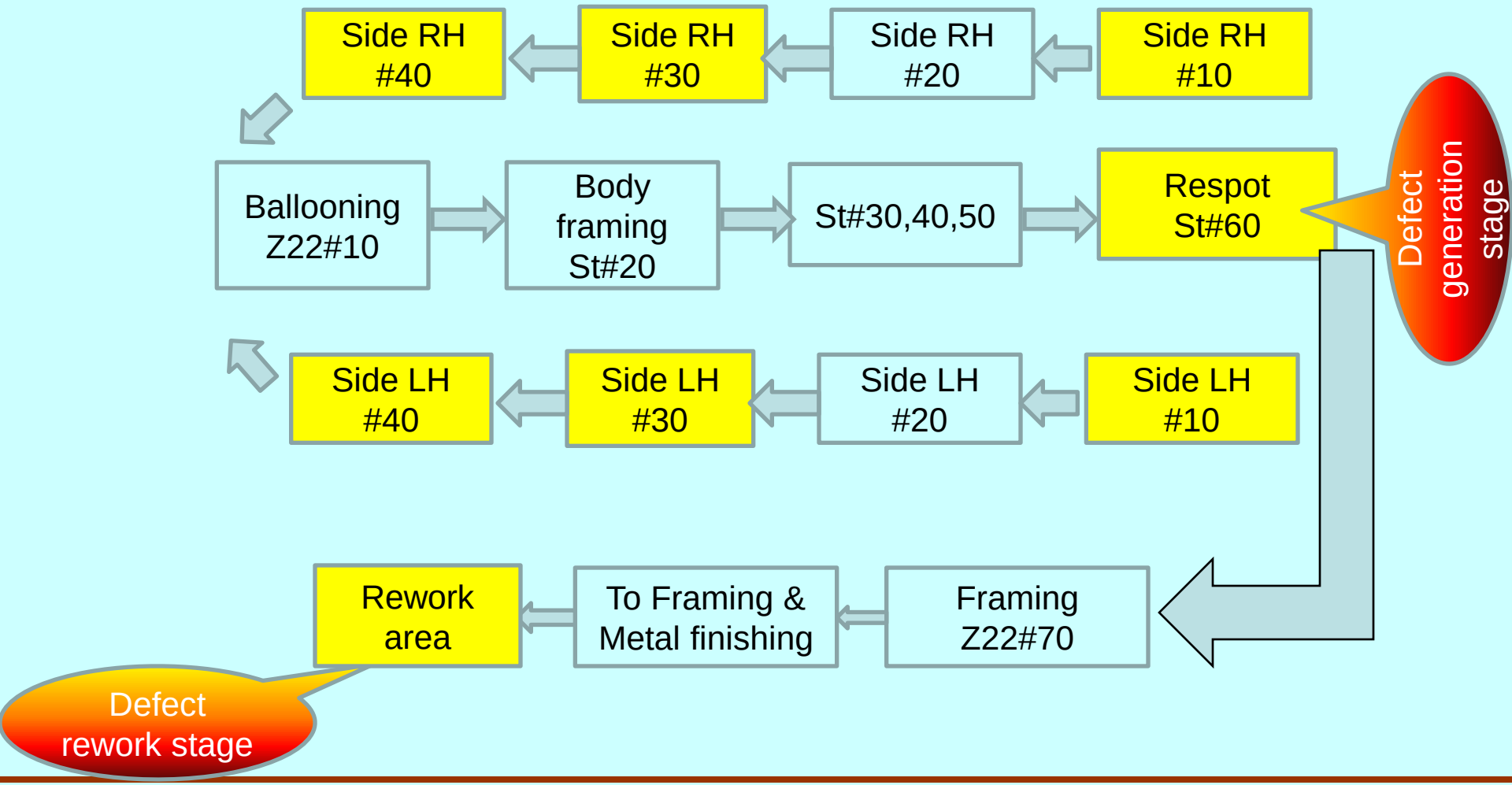
Countermeasure : Photograph & Details

डैमेज सारी गाडी व्यवस्थित चेक करणे व डैमेज हेवी डैमेज असल्यास ट्रेल्हल करिवर तशी नोंद करणे.

Positive Communication	Defect	Countermeasure
Date	26/6/12	
BWT Leader		
Trainee		

3. Understand the current situation

Flow chart



3. Understand the current situation

Process Flow

Side RH/LH#10

Side RH/LH#20

Side RH/LH#30

Side RH/LH#40

3. Understand the current situation

Observation of process

Side RH/LH#10

At Stn #10 while transferring the side from Stn #10 to Stn #20,
Cu plate mounting(hinge) Hitting to the side due to the sliding of side From rubber roller during transfer.

3. Understand the current situation

Observation of process

Side RH/LH#20

Side getting damaged while Transferring the side from Stn #10 to #20, #20 to #30, #30 to #40 due to the sliding of side during transfer through rubber roller.

3. Understand the current situation

Observation of process

Side RH/LH#30

While transferring
the side from Stn #30 to # 40,
Stn #30 guard
Is touching to the
side assembly

3. Understand the current situation

Observation of process

While transferring the side from Stn #30 to #40, side is coming contact with lifter edges.

3. Understand the current situation

Observation of process

Side RH/LH#40

While transferring the side From Stn #30 to #40, side is coming contact with Metal clamp.

3. Understand the current situation

Observation of process

While taking spot at respot#60, gun is hitting to tail lamp or W area.

4. Select Target

Present Level:

67 incidences FRC and 17 BIW's lost at FTBO reported In last 3 months

Target:

FTBO – ZERO Incidences

FRC : 80% Reduction of tail lamp area damage on framing line based on data of last 3 months

5. Analysis

Brainstorming by the CFT

Sr No	Observation	Possible cause
1	Observations related to handling issue	1.while transferring side assy from stn#10 to Stn#40 side hitting metal clamps
		2.while transferring side assy from stn#10 to Stn#40 side hitting metal parts of lifter
		3.while transferring side assy from stn#20 to Stn#30 side hitting electrical box
		4. due to large gap between roller side bumps while transferring

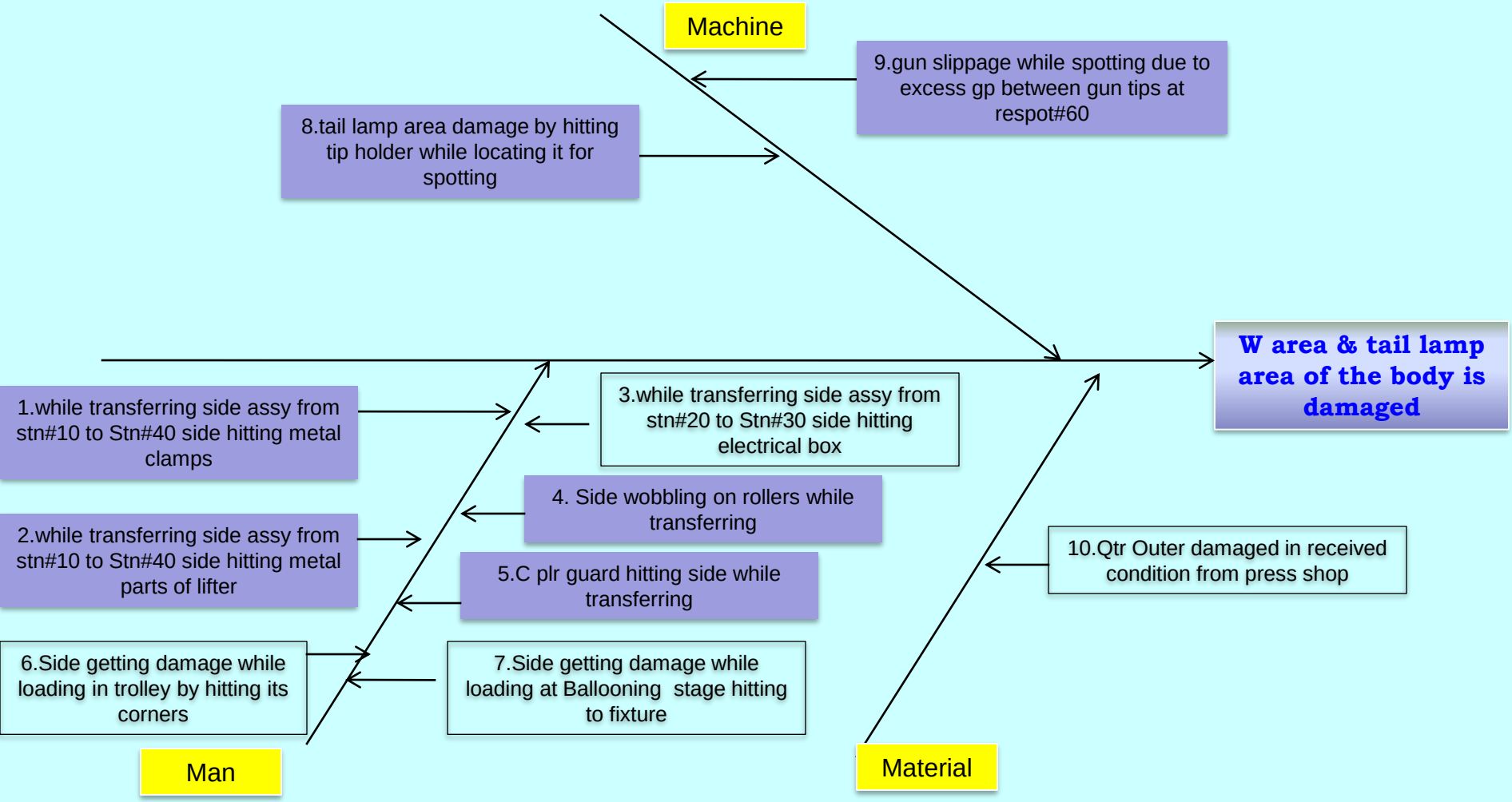
5. Analysis

Brainstorming by the CFT

Sr No	Observation	Possible cause
1	Observations related to handling issue	5.C plr guard hitting side while transferring
		6.Side getting damage while loading in trolley by hitting its corners
		7.Side getting damage while loading at Ballooning stage hitting to fixture
2	Spot welding	8.tail lamp area damage by hitting tip holder while locating it for spotting
		9.gun slippage while spotting due to excess gp between gun tips at respot#60

5. Analysis

5.1 Identify probable causes:



TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
1	While transferring side assy from stn#10 to Stn#40 side hitting metal clamps	At #40 Side is hitting to metal clamp during differential manual force of transferring the side assy	**Valid**
2	While transferring side assy from stn#10 to Stn#40 side hitting metal parts of lifter	While transferring from #30 to #40, side is hitting to Corners of the lifter	**Valid**
3	While transferring side assy from stn#20 to Stn#30 side hitting electrical box	Sufficient gap is observed .	In valid
4	Side wobbling on rollers while transferring	While transferring from #10 to #40, side is wobbling and due to that damages are happened	**Valid**
5	C plr guard hitting side while transferring	While transferring from #30 to #40, side is hitting to C plr guard metal during differential manual force of transferring the side assy	**Valid**
6	Side getting damage while loading in trolley by hitting its corners	Trolley corners has rollers .	In Valid

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
7	Side getting damage while loading at Ballooning stage hitting to fixture	Sufficient gap is observed .	In Valid
8	Tail lamp area damage by hitting tip holder while locating it for spotting	While locating the gun for spotting at respot #60 gun is hitting to outer surface of body causing damage at tail lamp area	**Valid**
9	Gun slippage while spotting due to excess gap between gun tips at respot#60	While locating the gun for spotting at respot #60, if gun opening is more due to tip worn out gun is hitting to outer surface of body causing damage at tail lamp area	**Valid**
10	Qtr Outer damaged in received condition from press shop	Checked pallet condition and received material found ok.	In Valid

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
1	While transferring side assy from stn#10 to Stn#40 side hitting metal clamps	At #40 Side is hitting to metal clamp during differential manual force of transferring the side assy	**Valid**

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
2	While transferring side assy from stn#10 to Stn#40 side hitting metal parts of lifter	While transferring from #30 to #40, side is hitting to Corners of the lifter	**Valid**

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
4	Side wobbling on rollers while transferring	While transferring from #10 to #40, side is wobbling and due to that damages are happened	**Valid**

Side transferring through rollers

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
5	C plr guard hitting side while transferring	While transferring from #30 to #40, side is hitting to C plr guard metal during differential manual force of transferring the side assy	**Valid**

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
8	Tail lamp area damage by hitting tip holder while locating it for spotting	While locating the gun for spotting at respot #60 gun is hitting to outer surface of body causing damage at tail lamp area	**Valid**

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
9	Gun slippage while spotting due to excess gp between gun tips at respot#60	While locating the gun for spotting at respot #60, if gun opening is more due to tip worn out gun is hitting to outer surface of body causing damage at tail lamp area	**Valid**

WHY - WHY Analysis

5.3 Identify the root causes:

Sr No	Probable cause	why	why	why	Action plan
1	Hitting metal clamps while transferring side assy from stn#10 to Stn#40 side	Differential manual force	No side guard for support from up side during transfer.	Design constraint – for spot welding.	Clamp position to be shifted.
2	Hitting metal parts of lifter while transferring side assy from stn#10 to Stn#40 side	Differential manual force	No side guard to support from upper side during side transfer.	Design constraint – for spot welding.	Additional rollers to be provided at the end of lifter bridge(#40)
3	Side wobbling on rollers while transferring	Height difference between rollers	Improper alignment between rollers among the stations		Roller distance is to reduced & leveling to be done
4	C plr guard hitting side while transferring	Differential manual force	No side guard for support from upper side during side transfer.	Design constraint – for spot welding.	To overcome guard height 2 additional rollers to be provided
5	Tail lamp area damage by hitting tip holder while locating it for spotting	Operator error	No gun guide		Guard to be provided
6	Gun slippage while spotting at respot#60	Excess gap between gun tips	Tip worn out while spotting		Guard to be provided

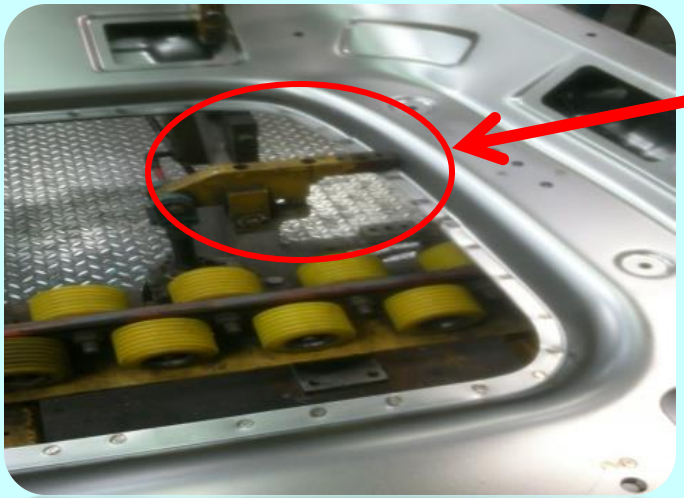
6. Implement the corrective measures

6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
1. Hitting metal clamps while transferring side assy from stn#10 to Stn#40 side	Clamp position shifted from D plr area to window area.	14/7/12 23/9/12	A R Shaikh	Completed

Before

After



6. Implement the corrective measures

6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
2. Hitting metal parts of lifter while transferring side assy from stn#10 to Stn#40 side	1. 4 Rollers to be provided at the end of lifter bridge(#40)(RH) 2. Fixture height near “Valli” Cu plate increased by adding 5 mm washer.	28/7/12 20/9/12	A R Shaikh	Completed

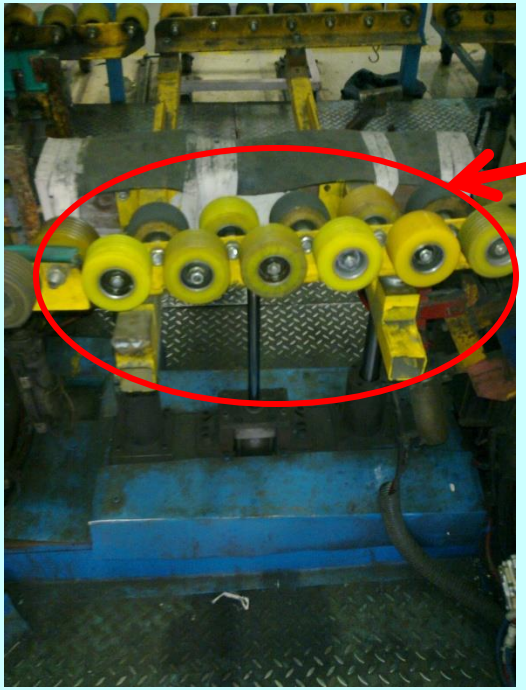
Provided 4 rollers at Side assy.

Roller height increased by adding 5 mm washer.

6. Implement the corrective measures

6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
4. Excess gap between transforming roller of side assy.	Roller pitch is to reduced(LH and RH) and Roller leveling to be done	4/8/12 11/8/12 18/8/12 25/8/12	A R Shaikh	Completed



Roller pitch is reduced from 60mm to 40mm & height increased upwards

6. Implement the corrective measures

6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
5.C plr guard hitting side while transferring	To overcome guard height additional rollers to be provided	22/8/12	A R Shaikh	Completed

Additional rollers provided

6. Implement the corrective measures

6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
8.Tail lamp area damage	Guard to be provided	18/7/12	A R Shaikh	Completed
9.Gun slippage while spotting at respot#60	Guard to be provided	15/9/12	A R Shaikh	Completed

Refer Attached Kaizen Sheet

Kaizen: Defect elimination of W-area spot damage

Safety/Ergo	Quality	Morale	Delivery/Stores	Cost	Productivity
	☆				

Date: July-12
Plant : Nasik
Origin :Scorpio body shop

Before

Spot damage at W-area due to human error

Description:

1. Excess gun movement due to spot welding on three plane with same gun.
2. Shank modification which use as gun guide not feasible.
3. W-area damage due to human error

After

Human error prevention by providing protector

Description:

1. Human error prevention by providing protector

Benefits

Tangible:

1. Elimination of defect W- area spot damage.

Intangible:

- Operator Morale Improved.

Reasons for No Recurrence of earlier state				
Applicable	✓		**Not Applicable**	

1. Nylon guard checking- Weekly

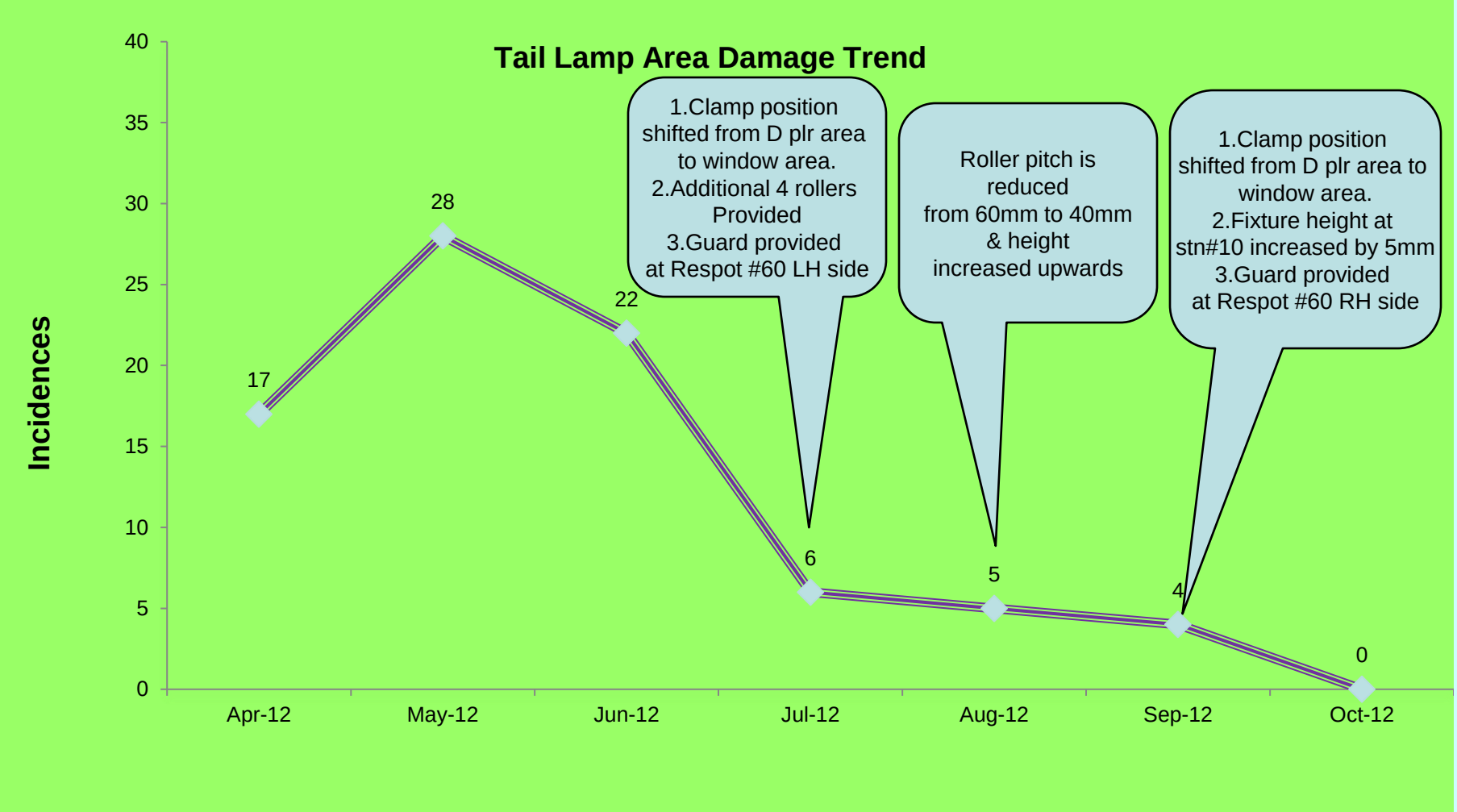
Cross Deployment Opportunity			
As Is	✓	Within Plant	✓
Concept		Across Plants	✓

	KDW	NSK	IGT	HDW	ZHB	Nasik II
Timeline	NA	July-12	NA	NA	NA	NA
Owner		**KD**				

Shared with	
All PU's - Nsk	✓

7. Check & confirm the results & effects

7.1 Verify Results:



7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

Fixture drawing updation to be done for -

1. Side #40 clamp position change, and
2. Respot #60 guard and gun guide

TDR already raised- TDR No- PE/TDR/F13/457

Check & confirm the results & effects (Cont……)

7.3 Tangible benefits:

- Improvement in FTBO by 0.1 %
- COPQ reduced by Rs 57780/- per annum
- FRC improvement at Body buyoff by 0.55%
- ZERO incidences reported at CAI i.e. downstream customer
- Reduction of the tail lamp area damage by 95% for the month Oct 12

Check & confirm the results & effects (Cont……)

7.3 Intangible benefits:

- Developed systematic approach for any chronic concern resolution
- Team work and confidence enhanced by resolving such chronic issue
- Hands on experience developed by implementing new pneumatic guard
- Alternative thinking by changing clamp location to avoid damage

8. Standardisation

8.1 Updation of documents:

1.Fixture drawing updation to be done for –
   a. Side #40 clamp position change, and
   b. Respot #60 guard and gun guide provided
2.Updated the fixture history card for said changes
3.Updated the fixture PM check sheet

FIXTURE HISTORY CARD

A	B	C	D	E	F
FIXTURE HISTORY CARD					
ZONE:	Z22				DEPT : SCORPIO BODY
FIXTURE	Z22-60		FREQUENCY : 6 Months and/or whenever changes done		
ASSY NA	Main framing 60				
DATE	UNIT NO	ITEM NO.	CONCERNS	CHANGES DONE/ ACTION TAKEN	REMARKS
27/05/08			Nut damaged due to spot taken on fender mtg nut	Nylon Gun guard provided	Fitment issue at Closure line.
18/7/12			Tail area damage while taking spot LH Side	Nylon gun guard provided on cylinder mounting.	i4-114644
15/9/12			Tail area damage while taking spot RH Side	Nylon gun guard provided on cylinder mounting.	FTBO Concern

FIXTURE HISTORY CARD

FIXTURE HISTORY CARD					
ZONE:	Z12				DEPT : MFG ENGG.-SCORPIO BODY
FIXTURE NO.	Z12-40		FREQUENCY : 6 Months and/or whenever changes done		
ASSY NAME:	SIDE RH 40				
DATE	UNIT NO	ITEM NO.	CONCERNS	CHANGES DONE/ ACTION TAKEN	REMARKS
6/4/08			Side assly scrapage	Guard height increased	
5/7/08			B Plr damage in between two nylon guards.	Single nylon gun guard provided.	
14/8/09			Scratch mark on B Plr	Sharp edge of mounting bolt removed.	QCRT
3/11/'09			Rocker panel Damage	Nylon block provided on side resting stopper.	QCRT CA3P0042/09
11/8/10			Damage B plr RH	Sharp edges of Side lifting tackle removed, Kinifoam applied	QCRT CA3P0073/10
28/6/11			Spot damage at C plr	Metalic stopper provided to restrict gun take spot on wrong location.	CA2P401/12
11/13/'11			Damage C Pillar RHS	Pipe insulation provided on lifter angle	CA3P122
28/07/12			Damage- Tail lamp area	4 Additional rollers provided in between stn. #30 and Stn #40 bridge	FTBO Concern
23/09/12			Damage- Tail lamp area	Clamp position shifted from D plr outer to window inside area.	FTBO Concern

Page 1

TOOL DESIGN REQUEST

Tool Design Request-PE - Microsoft Internet Explorer provided by Mahindra & Mahindra Ltd.

http://mconnect/sites/Nashik/Lists/Tool%20Design%20ReviewPE/History.aspx?SortField=CIR%5Fx0020%5Fer

File Edit View Favorites Tools Help

Favorites | Suggested Sites | Web Slice Gallery

Tool Design Request-PE

Page Safety Tools

PE/TDR/F13/364	SCORPIO BODY SHOP	7/11/2012 5:38 PM			7/19/2012	BHAT VIVEK	VARADE SACHIN	Approvec
	To reduce the defect of accelator bkt pedal mismatch on dash assly	10/9/2012 2:46 PM			10/11/2012	BHAT VIVEK	VARADE SACHIN	Approvec
PE/TDR/F13/457	Tail lamp spot damage in Scorpio body	10/11/2012 4:14 PM			10/15/2012	BHAT VIVEK	VARADE SACHIN	Approvec
PE/TDR/F13/476	Head lamp bkt damage in scorpio body shop	10/23/2012 11:20 AM			10/29/2012	BHAT VIVEK	VARADE SACHIN	Approvec
PE/TDR/F13/477	Spot damage at front floor #20 in scdc	10/23/2012 11:34 AM			10/29/2012	BHAT VIVEK	VARADE SACHIN	Approvec
PE/TDR/F13/446	CCB mtg unit on plennum modification in Scorpoi body	9/28/2012 4:31 PM	10/19/2012	11/2/2012	10/4/2012	BHAT VIVEK	VARADE SACHIN	Approvec
PE/TDR/F13/449	Back Door	10/8/2012 3:28 PM	10/19/2012	10/25/2012	10/12/2012	BHAT VIVEK	VARADE	Approvec

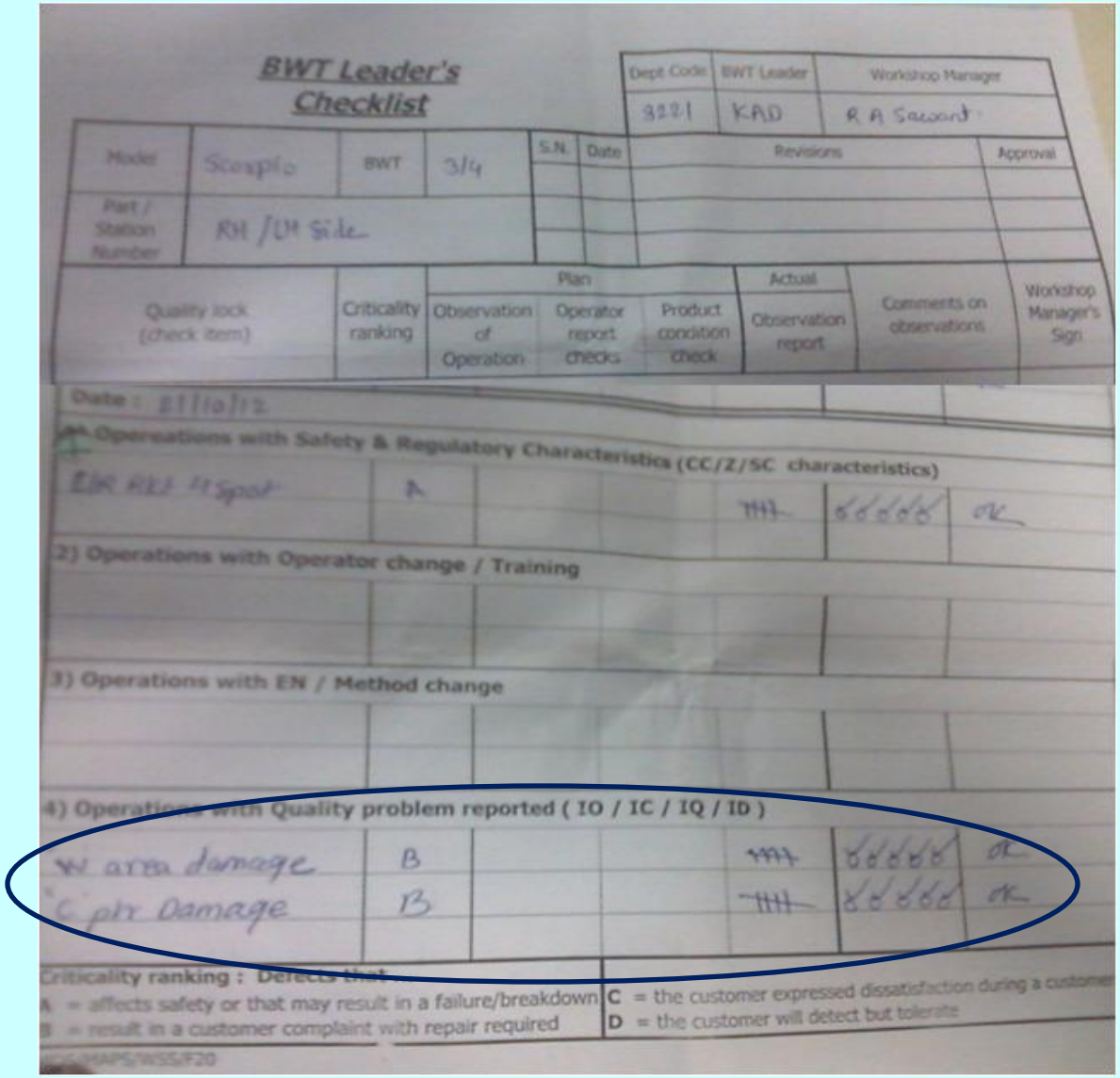
8.2 Training:

1. Operator were trained to handle the gun at Respot 60 to prevent spot damage and gun hitting.
2. Operator were trained to transfer the side from side assy. Stations with appx same forces.

8.3 Additional Audit / Check

1. Rubber rollers leveling and functional conditions of roller in PM.
2. Concern monitoring through Supervisory check sheet

Supervisory Check sheet



BWT Leader's Checklist

Dept Code	BWT Leader	Workshop Manager
3221	KAD	R A Sawant

Model	Scorpio	BWT	3/4	S.N.	Date	Revisions	Approval
Part / Station Number	RH / LH Side						

Quality lock (check item)	Criticality ranking	Plan: Observation of Operation	Plan: Operator report checks	Plan: Product condition check	Actual: Observation report	Comments on observations	Workshop Manager's Sign
Date : 21/10/12							
1) Operations with Safety & Regulatory Characteristics (CC/Z/SC characteristics)							
EIR RH 4 Spot	A			~~IIII~~	✓✓✓✓✓	OK	
2) Operations with Operator change / Training							
3) Operations with EN / Method change							
4) Operations with Quality problem reported (IO / IC / IQ / ID)							
W area damage	B			~~IIII~~	✓✓✓✓✓	OK	
C pkr Damage	B			~~IIII~~	✓✓✓✓✓	OK	

Criticality ranking : Defects that :

A = affects safety or that may result in a failure/breakdown
B = result in a customer complaint with repair required
C = the customer expressed dissatisfaction during a customer
D = the customer will detect but tolerate

MAPS/WSS/F20

9. Summarise & Plan future actions

9.1 Conclusion:

1.Damages happening at side assy are eliminated by changing the clamp position and doing the proper roller leveling of lifters for proper side transfer

2.Doing GEMBA with Team work leading to elimination of such chronic concern

9.2 Horizontal Deployment:

Project initiated to eliminate the B pillar area damage

9.3 Lessons Learnt:

Things Gone Wrong : Idea of Stopper provided on Respot Stn. #60 gun to eliminate the spot damage was not successful.

Things Gone Right : Problem can be solved easily by QC Story CFT Systematic problem solving approach

THANK YOU!!!